Kick-off Diagnostic · L2-4 — Overview of Complex Numbers

Grade 10

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (slides 2 and 3 of the pre-session must say this, so items 2 and 6 test it).
The one idea: $i = \sqrt{-1}$, so $i^2 = -1$; a square root of a negative number is written with i, and the powers of i repeat in a cycle of four.
The mistake most students make: Combining under one root: $\sqrt{-4} \cdot \sqrt{-9} = \sqrt{36} = 6$ instead of $2i \cdot 3i = -6$.
Readiness items 3, 4, 5, 7, 8 test: simplifying radicals ($\sqrt{48} = 4\sqrt{3}$), the coordinate plane, multiplying binomials, remainders on division by 4.
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All six stations, including the checks
B. I watched and read but skipped the checks
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

In the pre-session, what is the imaginary unit i?

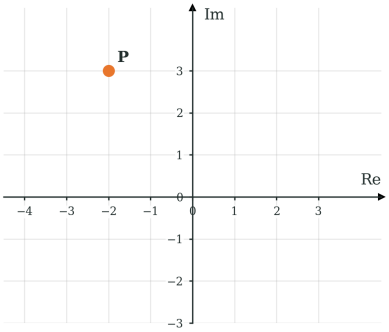
✓ **A.** $i = \sqrt{-1}$, **so** $i^2 = -1$
B. $i = -1$, so $i^2 = 1$
C. $i = \sqrt{1}$, so $i^2 = 1$
D. i is any unknown number

Signal. Wrong → did not watch station 1. Picking $i = -1$ (B) = mixes i with its square.

3 FILL IN THE BLANK · 30 s — READINESS

Point P on the complex plane represents _____ + _____ i

Accept: $-2 + 3i$



Signal. $3 - 2i$ → axes swapped; $2 + 3i$ → sign of the real part ignored. The $a + bi$ form is not yet secure.

4 CHECKBOX · 30 s — READINESS

Select the **two** true statements.

✓ **A.** $\sqrt{-9} = 3i$
B. $\sqrt{-9} = -3$
✓ **C.** $i^2 = -1$
D. $i^4 = -1$

Signal. Choosing $\sqrt{-9} = -3$ or $i^4 = -1$ → i is being treated as a sign, not a number.

5 MATCH · 30 s — READINESS

Match each power of i with its value.

i^2 → -1
i^3 → $-i$ (shown shuffled in Wayground)
i^4 → 1

Signal. Any miss → the cycle $i, -1, -i, 1$ is not known; today's powers of i depend on it.

6 MULTIPLE CHOICE · 30 s — **PRE-SESSION CHECK**

The pre-session warned about $\sqrt{-4} \cdot \sqrt{-9}$. Which wrong answer was it?

✓ **A.** 6 **(from** $\sqrt{36}$**)**
B. -6
C. $6i$
D. -36

Signal. The pre-session's "mistake most students make". Right idea, wrong warning → skimmed it.

7 REORDER · 30 s — **READINESS**

Put the steps for $(3+2i)(1-4i)$ in order.

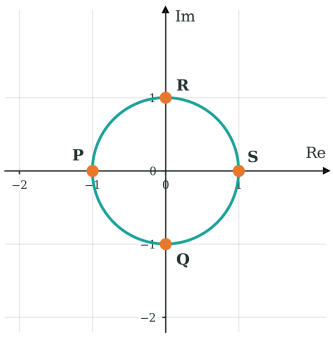
Correct order: $3-12i+2i-8i^2 \to 3-10i-8i^2 \to 3-10i+8 \to 11-10i$

Signal. Wrong → expanding, or replacing i^2 by -1, is the gap; expect -8 instead of $+8$.

8 DROPDOWN · 30 s — **READINESS**

Each point is a power of i on the unit circle. i^{43} is…

A. point P
✓ **B. point** Q
C. point R
D. point S



Signal. $43 = 4 \cdot 10 + 3$, so $i^{43} = i^3 = -i$. Point R (i) = used remainder 1; P = used i^2.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach radicals, $i^2 = -1$ and expanding with me. **3–4** → straight to the powers of i; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.